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Pretraining graph transformer for molecular representation with fusion of multimodal information[☆]

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ABSTRACT

Molecular representation learning (MRL) is essential in certain applications including drug discovery and life science. Despite advancements in multiview and multimodal learning in MRL, existing models have explored only a limited range of perspectives, and the fusion of different views and modalities in MRL remains underexplored. Besides, obtaining the geometric conformer of molecules is not feasible in many tasks due to the high computational cost. Designing a general-purpose pertaining model for MRL is worthwhile yet challenging. This paper proposes a novel graph Transformer pretraining framework with fusion of node and graph views, along with the 2D topology and 3D geometry modalities of molecules, called MolGT. This MolGT model integrates node-level and graph-level pretext tasks on 2D topology and 3D geometry, leveraging a customized modality-shared graph Transformer that has versatility regarding parameter efficiency and knowledge sharing across modalities. Moreover, MolGT can produce implicit 3D geometry by leveraging contrastive learning between 2D topological and 3D geometric modalities. We provide extensive experiments and in-depth analyses, verifying that MolGT can (1) indeed leverage multiview and multimodal information to represent molecules accurately, and (2) infer nearly identical results using 2D molecules without requiring the expensive computation of generating conformers. Code is available on GitHub¹.

1. Introduction

Molecular representation learning (MRL) has advanced the performance of assorted drug discovery tasks through self-supervised learning (SSL) and the pretraining and fine-tuning scheme [1]. Moreover, SSL facilitates efficient knowledge acquisition from limited labeled data by harnessing a substantial volume of unlabeled data. Notably, SSL has demonstrated remarkable achievements in the fields of natural language processing [2–4] and computer vision [5–7].

The recent advances of SSL in MRL have unveiled the potential of leveraging unsupervised pretraining models for drug discovery. Graph neural network (GNN) based MRL has achieved excellent performance for the tasks of molecular property prediction [8–16]. Transformer [17]-based MRL models have also presented promising

results for downstream tasks, such as GROVER [18], Uni-Mol [19], Graphormer [20,21] and Transformer-M [22].

However, pretraining on molecules remains a complex challenge. Pioneering work [8] has highlighted the necessity of training high-quality molecular representations at the node-level and graph-level. Relying on straightforward or coarse-grained alignment for molecules poses significant challenges in bridging the intrinsic correlation between modalities [10,11,16]. The adoption of a comprehensive representation of molecules beyond existing node and graph views is a crucial factor in enhancing the performance of MRL (Fig. 1), considering the multifaceted nature of molecular properties, which are influenced by diverse factors such as functional groups and 3D conformers [23–25]. Recently, multimodal MRL methods have emerged that leverage both 2D and 3D modalities within a unified framework [16,

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¹ <https://github.com/robbeplus/MolGT>

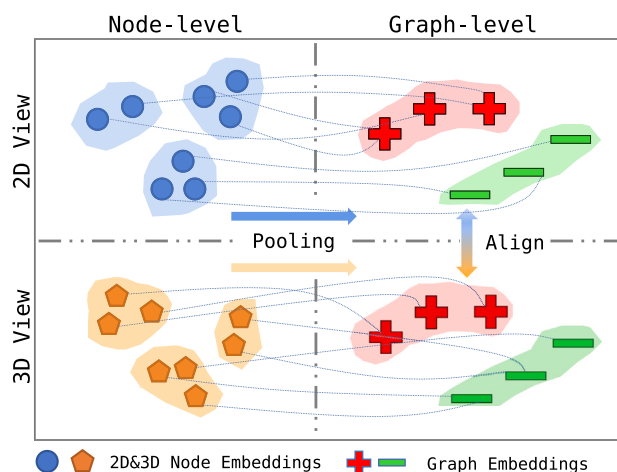


Fig. 1. Categorization of pretraining perspectives for MRL. To obtain high-quality embeddings at the node-level and graph-level, we extend the pretraining perspectives of the pioneering work [8] from solely node-graph tasks to include both 2D-3D and node-graph pretext tasks, leading to well-separated and aligned graph embeddings (denoted by their classes, + and -) between modalities using contrastive learning.

26]. Nevertheless, obtaining a 3D conformer of molecules may not be reliably available for certain tasks due to the substantial computational cost.

This paper proposes MolGT, a multiview and multimodal molecular pretraining framework that can perform fine-tuning without 3D conformers, which avoids the computational cost of generating 3D conformers. Our primary motivation is to pretrain MRL models that capture rich and explicit semantic information. To tackle this, we employ a customized modality-shared graph Transformer as the backbone model to encode different molecular modalities. Additionally, we integrate molecular fingerprints as a knowledge prior equipped with knowledge-guided prototypical clustering. Regarding node-level pretext tasks, InfoMotif and Denoising serve as complementary tasks to obtain high-quality embeddings at the node-level. Moreover, we adopt modality contrastive learning between molecular 2D topology and 3D geometry to encourage 2D networks to learn implicit 3D information. The pretext tasks collectively contribute to the comprehensive pretraining of our MolGT framework.

In summary, the main contributions are as follows:

- We propose an innovative modality-shared graph Transformer architecture specifically designed to handle 2D and 3D molecular data effectively. The architecture incorporates modality-specific modules, resulting in a parameter efficient framework that facilitates knowledge sharing between the two modalities.
- We propose MolGT, an SSL pretraining framework designed to facilitate joint pretraining on 2D and 3D molecules, including novel pretext tasks termed InfoMotif and knowledge-guided prototypical clustering. These tasks collectively contribute to the multiview and multimodal pretraining of the MolGT framework.
- We provide experimental analysis and insights with case studies to justify why the fusion of multiview and multimodal pretraining, along with the implicit 3D geometry, is beneficial.

2. Related work

2.1. Pretraining with 2D molecules

Recently, pretraining on molecular data has gained significant attention due to its ability to employ volumes of unlabeled data effectively. The pioneering work by Hu et al. [8] pretrained a five-layer

graph isomorphism network (GIN) [27] with node-level and graph-level tasks. Recent studies have adopted the contrastive paradigm with general graph augmentations for learning atom and molecule representations [9,28]. GROVER [18] is the first attempt to perform large-scale pretraining on molecular data using a graph Transformer. Mole-BERT [12] adopts a context-aware tokenizer to enlarge the atom vocabulary size and pretrain it with masked atoms modeling and contrastive learning. Moreover, numerous efforts have been made to leverage knowledge from external sources or other modalities, including knowledge graphs [29], cellular images [30], and reactions [31]. Clustering technique also has been adopted to enhance the representation learning on molecules [32].

2.2. Pretraining with 3D molecules

Most existing MRL models only assess molecules as 2D graphs, limiting their capacity to use 3D information for downstream tasks. Some research has started to incorporate 3D geometry to enhance biological data reliability. The existing 3D MRL models can be categorized into three distinct perspectives: explicit, mixture, and implicit, determined by whether they use 3D information during the inference stage.

2.2.1. Explicit 3D models

Many studies have demonstrated the potential of using molecular geometry to design networks that retain invariant and equivariant properties [33–35]. For example, GEM [34] explicitly leverages 3D information during the training and inference stages. Besides, coordinate denoising has emerged as a promising pretraining technique for 3D molecular geometry, exhibiting compelling potential across various drug discovery applications [19,36–38]. Explicit 3D models rely on numerous data regarding the positions of atoms, often generated using RDKit [39].

2.2.2. Mixture 3D models

Transformer-M [22] is a typical mixture 3D model. Transformer-M combines Transformer architecture with two learnable attention biases as specific modality channels. Transformer-M is capable of selectively activating specific channels based on the modality of the input data, allowing it to process and incorporate information from 2D and 3D modalities effectively. MoleBlend [16] is a multimodal pretraining case of Transformer-M, which integrates 2D and 3D modalities at atomic level.

2.2.3. Implicit 3D models

The goal of implicit 3D models is to enable a 2D model that can reason about 3D geometry from only 2D input. For instance, GraphMVP [10], MoleculeSDE [26] and 3D Infomax [11] inject 3D molecular geometry into a 2D molecular graph during pretraining. Typically, 3D Infomax uses multiple conformers that allow for knowledge transfer across different modalities of molecules. Further, GraphMVP proposes a multi-view pretraining framework, including contrastive and generative SSL. In addition, MoleculeSDE leverages group symmetric stochastic differential equation models to generate 2D and 3D modalities in the input space. When molecular data lack position information, implicit 3D models appear to be the optimal solution.

3. Preliminaries

A molecule can be represented as a graph G with discrete atom feature $\mathbf{H} \in \mathbb{R}^{n \times k}$, where $\mathbf{H} = \{h_1, \dots, h_n\}$, n denotes the number of atoms, and k represents the number of features. In addition, h_i can be converted into a dense feature vector $x_i \in \mathbb{R}^d$ through embedding layers, where d indicates the hidden dimension. For the 2D molecular topology, we define $G^{2D} = (\mathbf{H}, E)$, where $e_{ij} \in E$ denotes the edge feature between atoms i and j . For the 3D molecular geometry, we define $G^{3D} = (\mathbf{H}, R)$, where $R = \{r_1, \dots, r_n\}$, and $r_i \in \mathbb{R}^3$ represents the position of each atom.

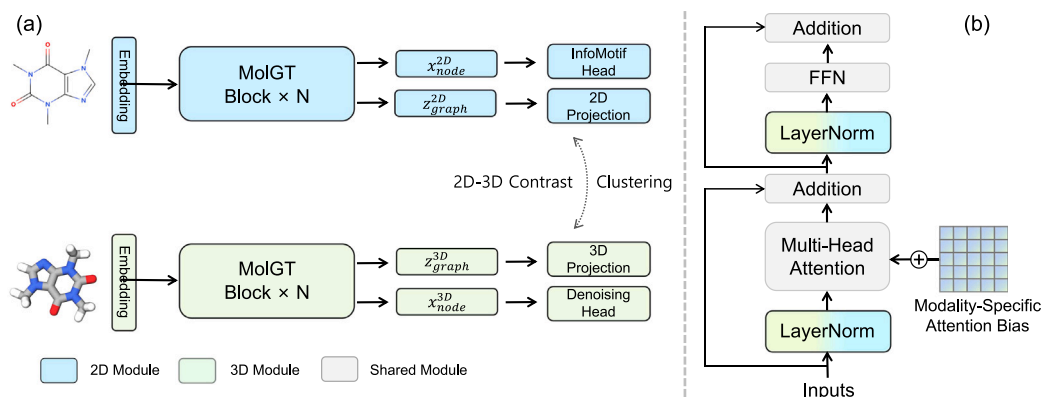


Fig. 2. Overview of the MolGT framework, demonstrating modality-specific components, including embedding layers, projection layers, layer normalization, and attention bias. (a): MolGT accepts 2D and 3D data and outputs node-level representation x_{node} and graph-level representation z_{graph} in 2D and 3D. These outputs are used in pretext tasks; (b): A MolGT block, each of which has modality-specific layer normalization, a shared-weights SelfAttention layer, and a feed-forward network layer.

3.1. Self-attention with bias

Given the representation pair x_i and x_j , the dot-product self-attention mechanism with learnable bias is computed as follows:

$$A_{ij} = \frac{(x_i W_Q)(x_j W_K)^T}{\sqrt{d}} + \Phi_{ij}, \quad (1)$$

where $x_i \in \mathbb{R}^{1 \times d}$ is projected by matrices $W_Q \in \mathbb{R}^{d \times d_Q}$ and $W_K \in \mathbb{R}^{d \times d_K}$ to the corresponding query and key. The dot-products of the queries and keys are divided by $\sqrt{d}$ to alleviate the gradient vanishing problem. In addition, A_{ij} is often called the attention matrix element, and Φ_{ij} is the learnable attention bias between atoms i and j .

3.2. Transformer-style architecture

A Transformer-style encoder primarily comprises a multi-head self-attention (SelfAttention) module and a position-wise feed-forward network (FFN) with layer normalization [40] and residual connection [41]. Following by PreLayerNorm [42], each Transformer encoder block can be written as follows:

$$X = \text{SelfAttention}(\text{LayerNorm}(X)) + X \quad (2)$$

$$X = \text{FFN}(\text{LayerNorm}(X)) + X. \quad (3)$$

4. The MolGT pretraining framework

In the following text, we present the detailed architecture of MolGT. Subsequently, we introduce a set of four pretext tasks designed explicitly for node view and graph view across both 2D and 3D modalities, as illustrated in Fig. 2(a).

4.1. Model architecture

The MolGT is built on Transformer-M [22] architecture consisting of a standard Transformer Encoder and learnable attention biases [20] for handling graph data. Let Φ^{2D} and Φ^{3D} denote the learnable attention biases, where Φ^{2D} consists of the distance of the shortest path encoding between a node pair and the average of the dot-products of the edge feature e_{ij} along the shortest path. In addition, Φ^{3D} is the Euclidean distance encoding with the Gaussian basis kernel function [21,43]. We selectively activate one bias based on the input modality. We have that:

$$A_{ij} = \frac{(x_i W_Q)(x_j W_K)^T}{\sqrt{d}} + \Phi_{ij}^{2D} \cdot \mathbb{I}_{2D} + \Phi_{ij}^{3D} \cdot \mathbb{I}_{3D}, \quad (4)$$

where $\mathbb{I}_{2D}$ and $\mathbb{I}_{3D}$ are indicator functions that activate the bias based on the input modality.

To enable parameter efficiency and facilitate knowledge sharing between modalities, we propose a tailored modality-shared graph Transformer. Recent works have suggested that Transformers can support learning across multiple modalities and allow knowledge sharing [44–47]. Inspired by the experimental analysis of parameter-sharing across modalities [44], we use three modality-specific modules including embedding layer, layer normalization [40], and projection layers, as illustrated in Fig. 2. In addition, MolGT uses modality-specific embedding layers to extract atom features, which are subsequently processed using a MolGT block. The MolGT block applies layer normalization before performing self-attention [42] with modality-specific attention bias(es) and FFN operations. Additionally, residual connections are employed around the self-attention and FFN layers.

4.2. SSL tasks for pretraining

4.2.1. InfoMotif

Subgraphs or substructures play a dominant role in graph representation [48]. Motifs [49] are significant subgraph patterns that frequently occur. The primary objective is to acquire distinct representations for each motif at the atom level to leverage the rich motif information. To derive a more fine-grained and atom-subgraph-level structural molecular understanding, we adopt a strategy of assigning positive pairs to atoms that belong to the same motif and negative pairs to atoms in different molecules, as depicted in Fig. 3(a). Hence, each motif can express distinct representations when appearing in different molecules. The training objective of InfoMotif $\mathcal{L}_{IM}$ for each atom i is given as follows:

$$\mathcal{L}_{IM} = -\log \frac{\sum_{j=1}^c \exp(f_{IM}(x_i, x_j))}{\sum_{k=1}^C \exp(f_{IM}(x_i, x_k))}, \quad (5)$$

where x_i denote the anchor atom representations, and x_j and x_k denote the representations of positive samples and negative samples, respectively. We randomly select c positive atoms and C negative atoms. Moreover, f_{IM} is the scoring function.

As mentioned above, the InfoMotif pretext task uses node-level contrastive learning to learn patterns pertaining to **functional groups**, **chemical bonds**, and **molecular topology**. While previous studies [50] have explored the influence of topology information in graph Transformers, they typically parallelize GNNs. In contrast, the InfoMotif objective can implicitly inject the context of topology by maximizing the mutual information between chemical structures. The attention map demonstrates that InfoMotif learns useful patterns in Section 6.3.

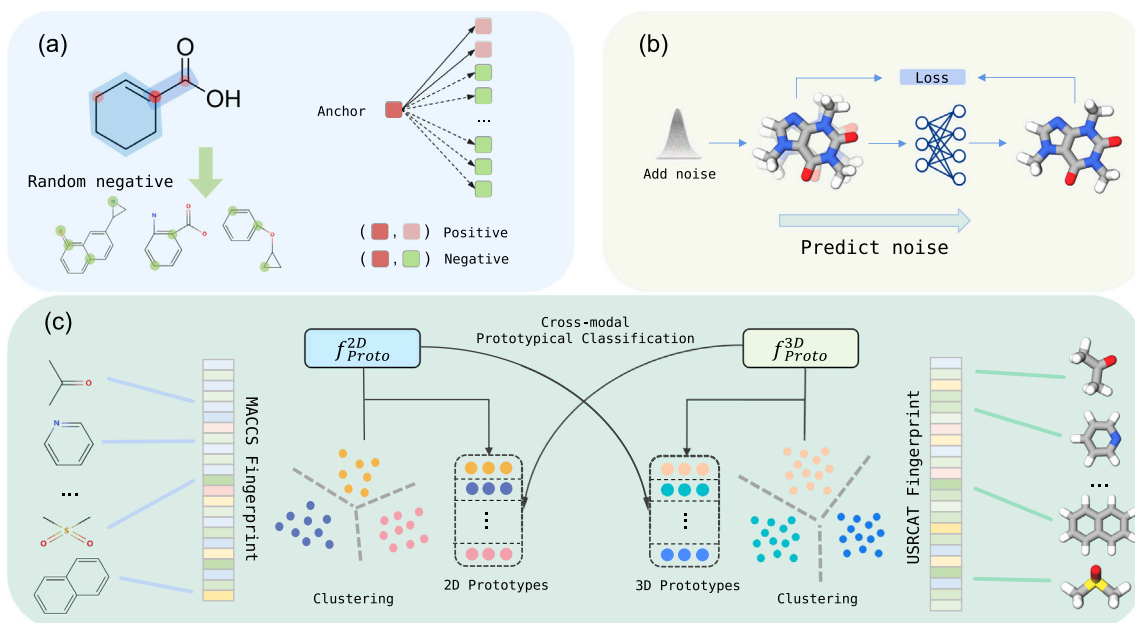


Fig. 3. Overview of self-supervised learning tasks. (a) InfoMotif for 2D node-level pretext task: The anchor atom is red in two motifs (the benzene ring and chemical bond). Negative atoms are randomly selected within a minibatch. (b) Coordinate denoising for 3D node-level pretext task. (c) Knowledge-guided prototypical clustering for 2D and 3D graph-level pretext tasks.

4.2.2. Coordinate denoising

The underlying principle of denoising aligns with the objective of learning the force field, as demonstrated in previous studies [36–38]. In the proposed approach, we incorporate 3D coordinate denoising as a pretext task designed explicitly for 3D node-level representations (Fig. 3(b)).

Formally, a noisy version of each position $\hat{r}_i = r_i + \epsilon$ is defined by adding noise ϵ . Then, a SE(3) equivariant attention layer [21] is adopted as the denoising (DN) head:

$$\hat{e}_i = \left(\sum_{j=1}^n A_{ij} \Delta_{ij} x_j W_{DN_2} \right) W_{DN_1}, \quad (6)$$

where x_j is the last hidden state of MolGT, A_{ij} denotes the attention score between atoms i and j calculated using Eq. (4) without ϕ^{2D} . Moreover, Δ_{ij} represents the directional vector $\frac{r_i - r_j}{\|r_i - r_j\|}$, and $W_{DN_2} \in \mathbb{R}^{d \times d}$, $W_{DN_1} \in \mathbb{R}^{d \times 1}$ are the learnable weight matrices.

We apply the Huber loss [51] as the denoising objective:

$$\mathcal{L}_{DN} = \begin{cases} 0.5 (\hat{e} - \epsilon)^2 / \delta & \text{if } |\hat{e} - \epsilon| < \delta \\ |\hat{e} - \epsilon| - 0.5 * \delta & \text{otherwise,} \end{cases} \quad (7)$$

where δ is the hyperparameter for the Huber loss.

4.2.3. Knowledge-guided prototypical clustering

Recent studies [52–54] have revealed that contrastive learning inherently facilitates the clustering of samples with respect to prototype-level patterns, enhancing downstream classification and compressing the data information. However, with the lack of explicit semantic information in molecular data, such prototypes cannot be attained via contrastive learning [55]. To address this limitation, we introduce Knowledge-Guided Prototypical Clustering (KGPC), as depicted in Fig. 3(c).

Molecular fingerprints (FP), which are essential descriptors containing comprehensive substructure and geometric information, can significantly enhance the performance of MRL models [56]. Following [54], we first construct prototypes from two fingerprints: MACCS [57] and USRCAT [58,59]. Specifically, we use K-Means clustering to generate prototypical labels for M molecules. We minimize the K-Means objective as follows:

$$\arg \min_{C^{2D}, C^{3D}} \underbrace{\sum_{k=1}^K \sum_{i=1}^M \|F P_i^{2D} - c_k^{2D}\|^2}_{\text{clustering 2D representations}} + \underbrace{\sum_{k=1}^K \sum_{i=1}^M \|F P_i^{3D} - c_k^{3D}\|^2}_{\text{clustering 3D representations}}, \quad (8)$$

where $F P^{2D}$ and $F P^{3D}$ denote MACCS and USRCAT respectively. After running K-Means clustering on fingerprints, we obtain the prototypical labels $y_i^{2D} \in \mathbb{R}^{K \times 1}$ and $y_i^{3D} \in \mathbb{R}^{K \times 1}$. To acquire cluster centers for each prototypical label, we pool the representations from multi-layer perceptron (MLP) projection heads f_{Proto}^{2D} and $f_{Proto}^{3D} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ on graph-level z_i^{2D} and $z_i^{3D} \in \mathbb{R}^{d \times 1}$ respectively. Thus, we obtain prototypes $C^{2D} \in \mathbb{R}^{K \times d} = [c_1^{2D}, c_2^{2D}, \dots, c_K^{2D}]$ and $C^{3D} \in \mathbb{R}^{K \times d} = [c_1^{3D}, c_2^{3D}, \dots, c_K^{3D}]$. The pooling operation can be expressed as follows:

$$c_i = \frac{1}{K_i} \sum_{i=1}^{K_i} f_{Proto}(z_i), \quad (9)$$

where c_i denotes the prototypical center from the MLP projection heads, and K_i represents the number of current labels.

The graph-level representations are expected to be sufficiently clustered in feature space, which is attributed to the prior structure and geometry knowledge from MACCS and USRCAT fingerprints. Prototypical clustering also encourages contrastive learning to maintain the alignment and uniformity of the feature distributions on the output unit hypersphere. We apply cross-modal prototypical classification [54] objective:

$$\mathcal{L}_{Proto} = - \underbrace{\sum_{i=1}^M \sum_{k=1}^K y_{i,k}^{3D} \log(p_{i,k}^{2D})}_{2D \text{ to } 3D} - \underbrace{\sum_{i=1}^M \sum_{k=1}^K y_{i,k}^{2D} \log(p_{i,k}^{3D})}_{3D \text{ to } 2D}, \quad (10)$$

where $p_{i,k}^{2D}$ and $p_{i,k}^{3D}$ are calculated using softmax:

$$\begin{aligned} p_{i,k}^{2D} &= \text{softmax}(C^{3D} \cdot f_{Proto}^{2D}(z_i^{2D})) \\ p_{i,k}^{3D} &= \text{softmax}(C^{2D} \cdot f_{Proto}^{3D}(z_i^{3D})) \end{aligned} \quad (11)$$

After the prototypical clustering objective converges, 2D and 3D graph-level representations gain sufficient knowledge from fingerprints.

Then, we replace the fingerprints with the representations from the MLP projection after the initial epochs. Thus, the new K-Means objective is as follows:

$$\arg \min_{C^{2D}, C^{3D}} \underbrace{\sum_{k=1}^K \sum_{i=1}^M \|f_{Proto}^{2D}(z_i^{2D}) - c_k^{2D}\|^2}_{\text{clustering 2D representations}} + \underbrace{\sum_{k=1}^K \sum_{i=1}^M \|f_{Proto}^{3D}(z_i^{3D}) - c_k^{3D}\|^2}_{\text{clustering 3D representations}}. \quad (12)$$

We suggest that robust alignment and uniformity can be achieved using explicit semantic information in concurrent contrastive learning. We leave the details on KGPC in Section 5.2.3.

4.2.4. Implicit 3D geometry

The goal is to align 2D and 3D modalities in the joint feature space by maximizing the mutual information. We consider 2D and 3D pairs for the same molecule as positive, and negative otherwise. Thus, the 2D molecular representation z^{2D} can align its positive geometric representation z^{3D} and contrast the N negative pairs within a minibatch simultaneously. For this purpose, we use InfoNCE (Noise-Contrastive Estimation) [60] as the contrastive loss:

$$\mathcal{L}_{\text{InfoNCE}} = - \left(\frac{1}{N} \log \sum_{i=1}^N \frac{\exp(\text{sim}(z_i^{2D}, z_i^{3D})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_i^{2D}, z_j^{3D})/\tau)} + \frac{1}{N} \log \sum_{i=1}^N \frac{\exp(\text{sim}(z_i^{3D}, z_i^{2D})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(z_i^{3D}, z_j^{2D})/\tau)} \right) / 2, \quad (13)$$

where sim denotes the dot product, and τ is a temperature hyperparameter. This enables MolGT to produce implicit 3D geometry for downstream tasks. Finally, the pretext tasks for the 2D and 3D node-level and 2D and 3D graph-level representations constitute a unified molecular pretraining framework. We use different weights α_i to balance the four losses, ensuring they are on a relatively similar numerical level. We minimize the following complete objective for MolGT:

$$\mathcal{L}_{\text{MolGT}} = \alpha_1 \mathcal{L}_{\text{IM}} + \alpha_2 \mathcal{L}_{\text{DN}} + \alpha_3 \mathcal{L}_{\text{Proto}} + \alpha_4 \mathcal{L}_{\text{InfoNCE}}. \quad (14)$$

5. Experiments

This section introduces the data and model architecture for pretraining. We conduct experiments on 12 molecular property prediction tasks, including eight classification datasets from the MoleculeNet benchmark [61] and four regression datasets. We also conduct experiments of quantum properties prediction on QM9 dataset. Finally, we perform a series of ablation studies to illustrate the benefits gleaned from the pretext task.

5.1. Pretraining

5.1.1. Datasets

We use three publicly available datasets containing 3D conformers for pretraining, with diverse molecule sizes and numbers of conformers. The pretraining datasets include **GEOM-Drugs** [62], **QMugs** [63] and **PCQM4Mv2** [64]. We select the few with the lowest energy when a molecule has multiple 3D conformers. We totally collect about 6M conformers.

5.1.2. Fingerprints

We utilize four publicly available datasets containing 3D information and generate 2D MACCS and 3D USRCAT fingerprints, as follows:

- **MACCS** [57]: The MACCS (Molecular ACcess System) keys are one of the most commonly used structural keys containing a subset of 166 keys. These 166 public keys are implemented in popular open-source cheminformatics software packages, such as RDKit. The fragment definitions for the 166 MACCS keys can be found in this document.² We use RDKit's MACCSkeys to generate the MACCS fingerprint.
- **USRCAT** [58,59]: Shape similarity refers to the comparison of the 3D shapes of the molecules, based on the principle that if the shapes of molecules overlap well, they can be considered similar. An extension of Ultrafast Shape Recognition (USR) called USRCAT, has been developed that includes pharmacophoric information while retaining performance benefits. We use RDKit's GetUSRCAT to generate the USRCAT fingerprint. We note that some molecules with a small number of atoms may fail to generate a 3D USRCAT fingerprint.

5.2. Pretext tasks details

5.2.1. InfoMotif

For simplicity and generality, we extract non-ring chemical bonds and simple rings via RDKit's GetSymmSSSR function as motifs. During training, we randomly select two positive atoms and 250 negative atoms. We employ the dot-product as the scoring function.

5.2.2. Denoising

We use the denoising method introduced in [21,22]. We apply an SE(3) equivariant attention layer for denoising. The backbone is an invariant(a special case of equivariant) model, which encodes 3D molecules using the distance encoding between different atoms. Following large scale denoising work [19], we randomly adding uniform noise $\epsilon \sim U(0, 1)$ for 15% of the atoms.

5.2.3. Knowledge-guided prototypical clustering

To learn multi-granularity knowledge from fingerprints, we create different cluster centers. Specifically, we establish three distinct centers with sizes of 500, 1000, and 3000 respectively. Thus, we have three KGPC losses, and we sum all sublosses to calculate $\mathcal{L}_{\text{Proto}}$.

Additionally, we adopt the episodic training strategy, in which we randomly select $m \ll M$ samples (1M in pretraining) from the entire dataset to accommodate a large volume of training data. Thus, the computational cost of feature collection and running the K-Means algorithm is relatively lower than that of traversing the entire dataset at a time. We use the GPU implementation of the K-Means algorithm from Faiss³ to accelerate training. The average time for the feature extraction, clustering, and pretraining stages of MolGT with an atom number of 40 is 6, 1, and 30 min, respectively.

5.3. Settings

We employ an eight-layer graph Transformer based on Transformer-M [22] architecture but use three modality-specific modules including embedding layer, layer normalization, and projection layer. The numbers of hidden dimensions, dimensions of the feed-forward layer, and attention heads are 512, 768, and 64, respectively. We use a linear network for the projection heads and prediction heads. The proposed model is optimized using Adam [65] optimizer with a learning rate

² <https://github.com/rdkit/rdkit/blob/master/rdkit/Chem/MACCSkeys.py>.

³ <https://github.com/facebookresearch/faiss>.

Table 1

We report the mean (and standard deviation) ROC-AUC of 3 seeds with scaffold splitting. Best and second-best results are marked in **bold** and underline, respectively. ‘No pretrain’ means training from scratch. We borrow the reported results of baselines from papers [10,12,16,19,26].

Classification (ROC-AUC%, higher is better ↑)									
Datasets # Molecules	BBBP 2039	Bace 1513	Tox21 7831	ToxCast 8575	Sider 1427	ClinTox 1478	MUV 93087	HIV 41127	Average –
No pretrain	66.1 (1.5)	73.2 (1.6)	72.1 (0.7)	62.3 (0.7)	59.6 (1.2)	72.7 (1.2)	74.3 (2.8)	70.1 (2.1)	68.8
AttrMask [8]	65.2 (1.4)	77.8 (1.8)	75.1 (0.9)	63.3 (0.6)	60.5 (0.9)	73.5 (4.3)	75.8 (1.0)	75.3 (1.5)	70.81
ContextPred [8]	70.6 (1.5)	78.8 (1.2)	73.6 (0.3)	62.6 (0.6)	59.7 (1.8)	74.0 (3.4)	72.5 (1.5)	75.6 (1.0)	70.93
Mole-BERT [12]	71.9 (1.6)	80.8 (1.4)	76.8 (0.5)	64.3 (0.2)	62.8 (1.1)	78.9 (3.0)	78.6 (1.8)	78.2 (0.8)	74.04
GROVER-base [18]	70.0 (0.1)	82.6 (0.7)	74.3 (0.1)	65.4 (0.4)	64.8 (0.6)	81.2 (3.0)	67.3 (1.8)	62.5 (0.9)	71.01
GROVER-large [18]	69.5 (0.1)	81.0 (1.4)	73.5 (0.1)	65.3 (0.5)	65.4 (0.1)	76.2 (3.7)	67.3 (1.8)	68.2 (1.1)	70.80
GraphMVP-G [10]	70.8 (0.5)	79.3 (1.5)	75.9 (0.5)	63.1 (0.2)	60.2 (1.1)	79.1 (2.8)	77.7 (0.6)	76.0 (0.1)	72.76
GraphMVP-C [10]	72.4 (1.6)	81.2 (0.9)	74.4 (0.2)	63.1 (0.4)	63.9 (1.2)	77.5 (4.2)	75.0 (1.0)	77.0 (1.2)	73.07
3D Infomax [11]	69.1 (1.2)	78.6 (1.9)	74.5 (0.7)	63.5 (0.8)	56.8 (2.1)	62.7 (3.3)	76.2 (1.4)	76.1 (1.3)	69.69
MoleculeSDE(VE) [26]	73.2 (0.5)	80.4 (0.9)	76.5 (0.3)	65.2 (0.3)	59.6 (0.8)	86.6 (3.7)	79.0 (0.2)	78.5 (0.3)	74.98
MoleculeSDE(VP) [26]	71.8 (0.8)	79.5 (2.1)	<u>76.8 (0.3)</u>	65.0 (0.3)	60.8 (0.4)	87.0 (0.5)	80.9 (0.4)	78.8 (0.9)	75.07
MoleBlend [16]	73.0 (0.8)	83.7 (1.4)	77.8 (0.8)	<u>66.1 (0.0)</u>	64.9 (0.3)	<u>87.6 (0.7)</u>	77.2 (2.3)	<u>79.0(0.8)</u>	<u>76.16</u>
MolGT	73.7 (1.0)	84.5 (0.1)	75.8 (0.2)	66.4 (0.2)	65.4 (0.4)	88.9 (0.4)	78.9 (2.2)	79.3 (2.1)	76.61

Table 2

MolGT performance on molecular property prediction regression tasks. We borrowed all results from [19]. The Uni-Mol paper did not report the results of GROVER on regression tasks Malaria and CEP, which is why they are not included in Table 2.

Regression (RMSE, lower if better ↓)					
Datasets # Molecules	ESOL 1128	Lipo 4200	Malaria 9999	CEP 29978	Average –
No pretrain	1.207 (0.053)	0.852 (0.022)	1.199 (0.023)	1.292 (0.031)	1.1375
GROVER-base	0.983 (0.090)	0.817 (0.008)	–	–	–
GROVER-large	<u>0.895 (0.017)</u>	0.823 (0.010)	–	–	–
3D Infomax	1.087 (–)	0.706 (–)	1.121 (–)	<u>1.218 (–)</u>	1.0330
GraphMVP-G	1.064 (0.045)	0.691 (0.013)	1.106 (0.013)	1.228 (0.001)	1.0222
GraphMVP-C	1.029 (–)	<u>0.681 (–)</u>	1.097 (–)	1.224 (–)	1.0077
Mole-BERT	1.015 (0.030)	0.676 (0.017)	1.074 (0.009)	1.232 (0.009)	<u>0.9992</u>
MolGT	0.839 (0.006)	0.788 (0.013)	1.074 (0.001)	1.021 (0.007)	0.9305

of 5×10^{-5} . The weight decay is 0.001. All pretraining experiments are conducted on $2 \times \text{A40 GPUs}$. To accelerate the pretraining in the experiments, we use a strategy similar to that for BERT [2]. We initially pretrain MolGT on molecules comprising up to 40 atoms, covering the majority of the training steps. Subsequently, for the remaining steps, we shift our focus to the larger molecules, defined as those containing up to 128 atoms to learn the Φ^{2D} and Φ^{3D} biases.

5.4. Molecular property prediction

5.4.1. Datasets

We adopt eight widely used classification datasets from MoleculeNet [61], including BBBP [66], Bace [61], Tox21 [67], ToxCast [61], Sider [68], ClinTox [69], MUV [70], and HIV [71]. We also report the performance of the four regression datasets, including ESOL [72], Lipo [73], Malaria [74] and CEP [75]. We use scaffold splitting [76] and report the mean and standard deviation for the results of three random seeds.

5.4.2. Baselines

Eight methods are considered as baselines, including 2D models and implicit 3D models, to demonstrate the effectiveness of MolGT. AttrMask [8], ContextPred [8], and Mole-BERT [12] are GNN-based 2D models. GROVER is a Transformer-based 2D model. GraphMVP [10], 3D Infomax [11] and MoleculeSDE [26] are implicit 3D models. MoleBlend [16] is the mixture model with 2D topology only.

5.4.3. Results

Tables 1 and 2 report the experimental results of MolGT and the competitive baselines. The performance of MolGT is significantly better than that of the 2D models and implicit 3D models, indicating that MolGT can take better advantage of the multiview and multimodal information of the molecule.

Comparatively, MolGT outperforms the 2D GNN-based models and implicit 3D models in five classification datasets, including BBBP, Bace, ToxCast, Sider, and ClinTox. For regression tasks, MolGT outperforms the previous methods in three datasets, including ESOL, Malaria, and CEP. Further, MolGT results in a notable absolute reduction of the root mean squared error (RMSE) compared to the previous methods. This improvement over previous works is primarily due to the global modeling and implicit 3D information in MolGT.

Moreover, MolGT can outperform the 2D Transformer-based GROVER and MoleBlend. The proposed MolGT only contains $\sim 16\text{M}$ parameters whereas the GROVER-base and GROVER-large models have $\sim 48\text{M}$ and $\sim 100\text{M}$ parameters, respectively. However, MolGT does not exceed the state-of-the-art results on the Lipo and Tox21 datasets, a performance degradation also observed in the Transformer-based GROVER and MoleBlend, which might be attributable to the Transformer architecture [56].

5.5. Quantum properties prediction

5.5.1. Dataset

We use QM9 [77] dataset to evaluate the effectiveness of the implicit 3D geometry of MolGT on 3D tasks. The dataset contains around 130,000 small organic molecules and is widely used as a quantum property prediction benchmark [38]. Every molecule has the 3D coordinates of one low-energy conformer and is annotated with 12 quantum properties calculated by density functional theory (DFT) algorithm. We follow the experimental setting introduced in [11,38], and report eight quantum properties used in [11].

5.5.2. Baselines

For comparison, we use baselines introduced in [11], including GraphCL, ProPred, DisPred, and ConfGen. GraphCL is a conventional 2D pretraining method. ProPred refers to pretraining by predicting the

Table 3

Performance (MAE, lower is better) on QM9. Baselines' results are borrowed from [11,38]. Note that only GNS-TAT is pretrained with all atom conformers(including hydrogen atoms).

Models	μ	α	HOMO	LUMO	GAP	R^2	ZPVE	C_v
GraphCL	0.3937	0.3295	79.57	80.81	120.08	21.84	12.39	0.1422
ProPred	0.3975	0.3732	93.11	99.84	131.99	29.21	11.17	0.1795
DisPred	0.4626	0.3570	80.58	84.93	116.21	29.23	25.91	0.1587
ConfGen	0.3940	0.4219	79.75	79.16	110.72	20.86	21.10	0.1555
3D Infomax - Drugs	0.3512	0.2959	70.78	71.38	102.59	18.96	9.677	0.1409
3D Infomax - QMugs	0.3668	0.2807	70.77	78.10	103.85	18.00	12.06	0.1208
GNS-TAT (all atom, 3D)	0.016	0.040	14.9	14.7	22.0	0.44	1.018	0.020
MolGT	0.3777	0.2508	57.80	70.71	99.88	16.61	6.45	0.1056

Table 4

Ablation on classification tasks. Avg refers to the average ROC-AUC score across the six classification tasks.

Loss	2D-node	3D-node	Graph	Avg
No pretrain				67.67
CL			✓	68.52
KGPC+CL			✓	72.23
IM+DN	✓	✓		68.92
DN+KGPC+CL		✓	✓	75.67
IM+KGPC+CL	✓		✓	74.82
IM+DN+PC+CL	✓	✓	✓	73.17
IM+DN+KGPC+CL	✓	✓	✓	76.61

Gibbs free energy of GEOM-Drugs' pretraining subset. DisPred refers to predicting all atom distances, and ConfGen means pretraining by predicting up to 10 conformers. For a fair comparison, we use 3D Infomax pretrained on the GEOM-Drugs and QMugs datasets, denoted as 3D Infomax-Drugs and 3D Infomax-QMugs, respectively. We also compare with an all atom denoising model (including hydrogen atoms), GNS-TAT [38].

5.5.3. Results

We evaluate MolGT with 2D topology only. Table 3 confirms that the MolGT with implicit 3D geometry substantially improves quantum properties prediction, compared to contrastive learning on modalities only or conventional 2D models. However, the quantum properties in QM9 dataset are closely related to all atom conformers (including hydrogen atoms), rather than just heavy atoms (C, N, O, etc.). Note that all baselines, including MolGT, are pretrained using only heavy atoms (excluding hydrogen atoms). A discussion on this limitation is provided in the Appendix B.

5.6. Ablations study

We analyze the contributions of various SSL pretext tasks from three perspectives. We conduct ablation studies to confirm the effectiveness and validity of the proposed pretext tasks, by removing some components of MolGT. More ablation studies can be found in Appendix A. We abbreviate the terms InfoMotif loss, Denoising loss, Prototypical Clustering (without Knowledge-Guided) loss, Knowledge-Guided Prototypical Clustering loss, and contrastive learning loss as IM, DN, PC, KGPC and CL respectively, as illustrated in Table 4.

5.6.1. Effect of the node-graph view

To evaluate the effects of the node-graph view, we remove the node-level tasks or graph-level tasks. It becomes apparent that using a single-view pretext task yields limited performance. The model underperforms in downstream tasks when contrastive learning is absent. This observation substantiates the underlying hypothesis that the incorporation of both node-level and graph-level tasks significantly enhances the performance of the MRL model.

5.6.2. Effect of the 2D and 3D modalities

We study the effects of the pretext tasks on different modalities. Performing pretext tasks on a single modality can lead to diminished performance. Imbalanced learning objectives on one modality lead to the inequality of representations on another. This verifies our claim that adopting SSL at the 2D and 3D modalities is beneficial.

5.6.3. Effect of knowledge guidance

For the graph-level pretext task, we verify whether using the fingerprints as a knowledge prior can benefit the downstream task performance. Table 4 reveals that the KGPC technique outperforms the vanilla prototypical clustering method, as it encourages the formation of meaningful, graph-level knowledge-aware anchors. Because molecules or MRL models lack explicit semantic information, performing vanilla prototypical clustering results in randomly distributed semantic anchors, which offer no benefit to representations.

6. Case study

6.1. Does MolGT learn 3D information?

To explore whether MolGT can learn 3D Information, we conduct downstream tasks on MoleculeNet with two modes: activating the 3D channel and activating the 2D and 3D channels, where the 2D and 3D channels denote the use of 2D and 3D biases, respectively. We first generate five conformers for each molecule in the MoleculeNet datasets using RDKit. Then, we conduct classification experiments using the same settings and run them five times for each conformer. Fig. 4 summarizes the average results of the five experiments. When only using the 3D channel, the model still outperforms the baselines and exhibits nearly identical results compared to only using the 2D channel. Activating both 2D and 3D channels led to a drop in performance. This can be attributed to the mismatch in bias information between the pretraining stage, where only 2D or 3D biases were used, and the fine-tuning stage, where both were activated.

Using both 2D and 3D information for 3D molecules is unnecessary because 2D-related features are discontinuous and non-essential, especially for the denoising task [79]. The aforementioned analysis and facts imply that the implicit 3D model can achieve identical performance without computing the 3D conformer for each molecule before inference. We hope the implicit 3D models can accelerate large-scale biology and chemistry experiments for real-world applications.

6.2. Molecule clustering analysis

To gain deeper insight into the KGPC task, we visualize the molecular representations of MolGT to confirm whether it captures meaningful patterns of molecules (e.g., scaffold) or leads to explicit semantic clustering results. Following [80], We first select the six most common scaffolds from the ZINC [81] dataset and randomly sample 300 molecules for each selected scaffold. We use t-SNE [82] to visualize the representations from one pretraining without KGPC and one pretraining with KGPC. Fig. 5 compares MolGT without KGPC, revealing that

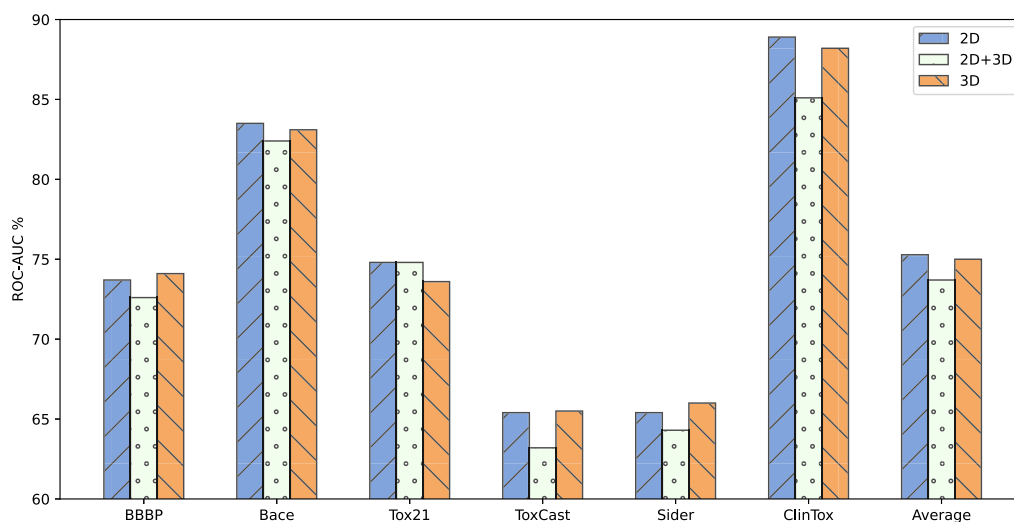


Fig. 4. Comparison of activated channel(s) on MoleculeNet.

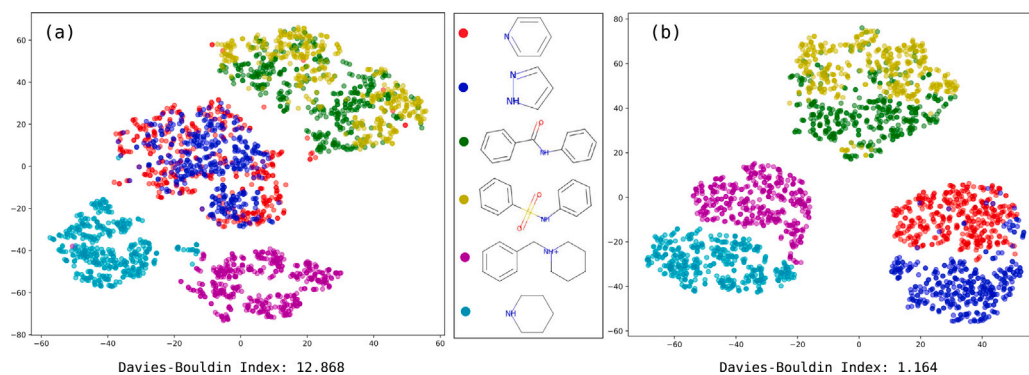


Fig. 5. Visualization of molecular representation, where different colors signify molecules with the different scaffold. (a): MolGT without KGPC is incapable of distinguishing similar molecules but has different scaffolds. (b): In contrast, MolGT with KGPC can encode scaffold information effectively. The Davies Bouldin index (DB index) [78] indicates the quality of clustering, where a lower DB index implies a more effective separation within the clustering.

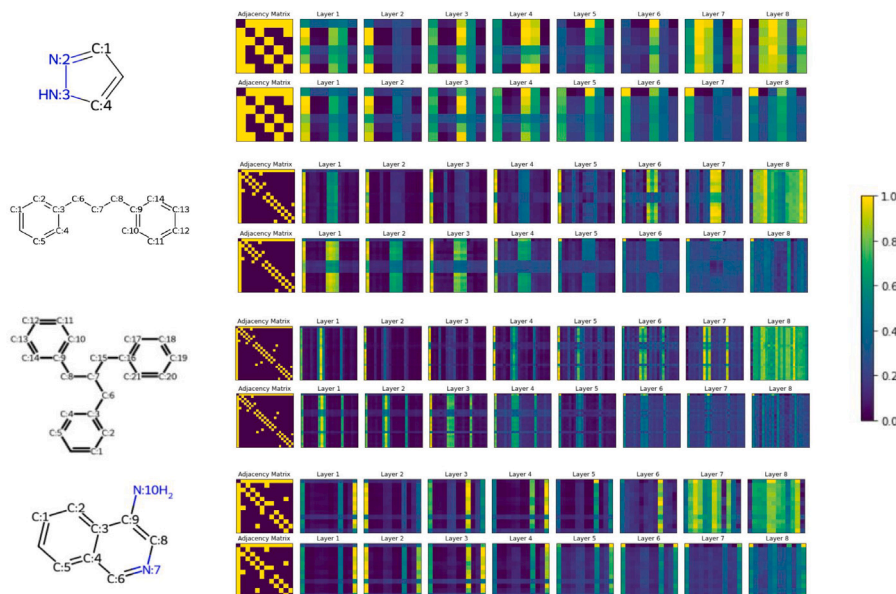


Fig. 6. Visualization of attention maps from four molecules with special and representative structures. Each row from left to right represents the original adjacency matrix and attention score matrices from lower to higher layers. For the adjacency matrix, yellow and dark purple indicate that the two nodes are connected and disconnected respectively. For the attention map, brighter yellow indicates higher attention and darker colors indicate lower attention. The first node in the attention map is the graph token. The first row for the same molecule represents the use of InfoMotif, whereas the second row indicates the absence of InfoMotif.

MolGT with this task has more-distinctive clusters. This visualization proves the claim that using fingerprints as a prior can create explicit semantic anchors. Moreover, these semantic anchors also benefit the expressive power of MolGT to capture and distinguish similar molecular structures.

6.3. Attention maps analysis

By visualizing the attention maps (Fig. 6), we can infer which features are most relevant to the model's decision-making process. Thus, all self-attention weights of the heads for each layer in the model are averaged together. We choose four representative molecules with distinctive substructures and topologies. For each molecule, we present a comparative visualization featuring two attention maps: one from a model trained with all pretext tasks and another from a model trained without InfoMotif. The first model can capture lower-layer features and substructures and prefers to make an average pooling on the top layer. The second model only captures lower-layer substructures and cannot aggregate atom features on higher layers. This phenomenon occurs as the model excessively concentrates its attention on the graph token on higher layers, losing the capability to learn helpful representations further. The InfoMotif pretext task effectively helps the model appropriately distribute attention across various substructures and aggregate features on higher layers.

7. Conclusions and future work

We propose MolGT, a comprehensive and complete MRL framework. We extend the MRL framework to incorporate both node and graph-level views and 2D and 3D modalities. The proposed MolGT with modality-specific modules exhibits parameter efficiency and facilitates knowledge sharing between the two modalities. To improve and perfect existing MRL pretext tasks, we propose two pretext tasks: InfoMotif for 2D node-level pretraining and Knowledge-Guided Prototypical Clustering for 2D and 3D graph-level pretraining. With the ability of implicit 3D geometry reasoning, MolGT achieves better performance than 2D and implicit 3D models on several molecular benchmark datasets.

Nevertheless, there is still considerable space for improving MolGT. For future work, a promising aspect would be to mine invariant subgraph learning inspired by causality to explore the out-of-distribution problem. By integrating invariant subgraph learning, MolGT could potentially achieve greater robustness and adaptability when faced with novel molecular structures and properties. In addition, a straightforward improvement is to scale up the model size used in MolGT and elaborate on more sophisticated graph Transformer encoders. Work using MolGT in more drug discovery applications would also be valuable, such as protein-related tasks.

CRediT authorship contribution statement

Ruizhe Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chunyan Li:** Writing – review & editing, Supervision, Methodology. **Longyue Wang:** Writing – review & editing, Supervision. **Mingquan Liu:** Writing – review & editing, Validation, Investigation, Data curation. **Shugao Chen:** Writing – review & editing, Validation, Data curation. **Jiahao Yang:** Writing – review & editing, Validation, Data curation. **Xiangxiang Zeng:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Extended ablation studies

A.1. Effect of pretraining data

The proposed MolGT model is pretrained using the GEOM-Drugs, QMugs, and PCQM4Mv2 datasets. For a fair comparison with MoleBlend, we retrain MolGT using only the PCQM4Mv2 dataset. As mentioned in Section 5.2.2, we adopt a strategy that randomly select 1M molecular for current training epoch. Therefore the amount of pre-training data is not sensitive to the performance of downstream tasks. Besides, the PCQM4Mv2 dataset provides high-quality equilibrium 3D conformers. The new version of MolGT, is denoted with “*” in the upper right corner. Table A.5 presents the results of classification tasks on MoleculeNet.

A.2. Effect of number of cluster centers

To determine the appropriate number of cluster centers (prototypes), we use the knee point method and follow the approach of previous clustering-based SSL work [83]. Specifically, the number of cluster centers at the knee point for 2D MACCS fingerprints and 3D USRCAT fingerprints are approximately 500 and 1000, respectively. Additionally, the number of cluster centers used in [83], which demonstrated the best performance, are 1000 and 3000, respectively. Based on these findings, we use 500, 1000, and 3000 as the number of cluster centers. Empirically, we find that using multiple centers can yield better performance. We evaluate the classification performance on BBBP, Base, Sider, and ClinTox dataset. The results are shown in Table A.6.

A.3. Effect of temperature parameter

The appropriate temperature parameter can stabilize contrastive learning. We use the common temperature parameters 0.07 and 0.1 introduced in [84], as well as 0.01 and 1.0. We find that the common temperature parameters yield similar results. However, using a temperature that is too high or too low results in poor performance. The results are shown in Table A.7.

Appendix B. Limitations

MolGT does have some limitations. For instance, it performs poorly on the QM9 dataset. This is because the quantum properties in QM9 are closely related to all atom conformers (including hydrogen atoms), rather than just heavy atoms (C, N, O, etc.). For example, in a simulation system, water and an oxygen atom are totally different, whereas they appear the same if we discard all hydrogen atoms. MolGT is pretrained using only heavy atoms, excluding hydrogen atoms. The limitation indicates that while MolGT is effective in some downstream tasks, it may not perform as well in scenarios where all atom information is crucial. We leave the exploration of an all atom implicit 3D model for future work.

Appendix C. Decomposition of molecules

We provide the pseudocode to better illustrate the process of motif decomposition, as shown in Algorithm 1. After the decomposition, an example molecule shown in Fig. C.7 is decomposed to the following motifs: [[0, 1], [1, 2], [2, 3], [3, 4], [2, 5], [5, 6], [5, 7], [9, 23], [7, 22, 21, 10, 9, 8], [11, 12, 20, 19, 21, 10], [13, 12, 20, 15, 14], [16, 17, 18, 19, 20, 15]]. We can observe that all atoms are decomposed to four rings and eight bonds.

Table A.5

Results on MoleculeNet classification tasks.

Classification (ROC-AUC%, higher is better ↑)									
Datasets	BBBP	Bace	Tox21	ToxCast	Sider	ClinTox	MUV	HIV	Average
# Molecules	2039	1513	7831	8575	1427	1478	93 087	41 127	–
MoleBlend	73.0 (0.8)	83.7 (1.4)	77.8 (0.8)	66.1 (0.0)	64.9 (0.3)	87.6 (0.7)	77.2 (2.3)	79.0(0.8)	76.16
MolGT	73.7 (1.0)	84.5 (0.1)	75.8 (0.2)	66.4 (0.2)	65.4 (0.4)	88.9 (0.4)	78.9 (2.2)	79.3 (2.1)	76.61
MolGT*	74.0 (1.2)	83.7 (0.2)	74.9 (0.1)	66.7 (0.3)	65.1 (0.2)	87.8 (0.5)	79.3 (1.7)	78.4 (2.6)	76.24

Table A.6

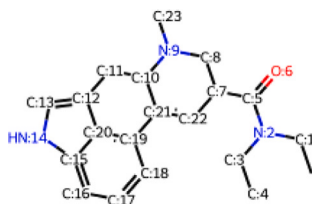
Average results on BBBP, Bace, Sider, and ClinTox classification tasks. ‘fused’ means using multiple centers.

Number of prototypes	500	1000	3000	Fused
Average ROC-AUC%	76.48	76.32	77.42	78.13

Table A.7

Average results on BBBP, Bace, Sider, and ClinTox classification tasks.

Temperature	0.01	0.07	0.1	1.0
Average ROC-AUC	74.32	78.01	78.13	75.92

**Fig. C.7.** SMILES: CCN(CC)C(=O)C1CN(C2CC3=CNC4=CC=CC(=C34)C2=C1)C.**Algorithm 1** Decomposition Algorithm**Input:** RDKit molecule *mol***Output:** list of cliques

```

cliques ← []
for each bond in mol.GetBonds() do
    a1 ← bond.GetBeginAtom().GetIdx()
    a2 ← bond.GetEndAtom().GetIdx()
    if not bond.IsInRing() then
        cliques.append([a1, a2])
    end if
end for
ssr ← [list(x) for x in Chem.GetSymmSSSR(mol)]
cliques.extend(ssr)

```

Appendix D. Discussion on modality sharing

We discuss the limitations associated with adopting the Asymmetric Dual Encoder (ADE) or Siamese Dual Encoder (SDE) architectures in MolGT. ADE architectures suffer from either partial or no parameter sharing, resulting in increased computational costs and the inability to share knowledge across modalities effectively. Conversely, SDE architectures share parameters between the two encoders, but using SDE architectures for MolGT in the context of contrastive learning tasks tends to lead to overfitting and collapse [85]. This limitation can be attributed to the challenge faced by encoders in learning meaningful representations when nearly all parameters, except biases, are shared.

We noticed that previous attempts to adopt contrastive learning for molecules at scale have not been successful under Transformer architecture. The previous works leveraged contrastive learning for molecules use two different GNN networks for 2D input and 3D input respectively [10,11]. In the case will not suffer the issue of mode collapse. The graph Transformer architecture is typically an unordered model without inductive bias, unless the use of attention bias(es)

between nodes. If we use a SDE to encode the same molecule with both 2D and 3D inputs, almost all outputs of the networks are identical except for the attention bias network. This could lead the network to learn shortcut solutions during 2D-3D contrastive learning.

We conduct two experiments to verify this hypothesis. First, we used the SDE architecture, set the output dimension to 128, and performed only 2D-3D contrastive learning. Then, we used the same settings but replaced the SDE with the modality-shared graph Transformer proposed in MolGT. We plotted the number of collapsed dimensions [85] and the accuracy of the contrastive learning task, as shown in Fig. D.8. The results show that the SDE experiment exhibits abnormally high accuracy, which is indicated by the high number of collapsed dimensions. This suggests that the representations from the model are in a relatively constant space [85]. In contrast, the modality-shared graph Transformer does not suffer from the dimension collapse issue and performs well in downstream tasks.

The exploration of additional architectures, such as the ADE, is beyond the scope of this paper. We adopt the modality-shared graph Transformer because it is a feasible architecture for performing contrastive learning and other pretext tasks.

Appendix E. Discussion on motivation

The primary contribution of MolGT is the proposed implicit 3D model for large-scale molecular representation tasks, which avoids the computational cost of generating 3D conformers. To achieve this, we leverage multiview of molecular data and integrate them effectively. Inspired by the success of the CLIP model [86] in vision-language contrastive learning, we aimed to explore whether 2D-3D contrastive learning could similarly enhance large scale molecular representation learning under Transformer architecture. However, previous attempts to adopt contrastive learning for molecules at scale have not been successful (see Table 18 in the Uni-Mol paper [19]). We believe this failure is due to the lack of explicit semantic information in molecular data. In vision-language contrastive learning, pretrained models can distinguish between objects like dogs and cats without explicit labels [52]. We pose the question: can contrastive pretraining on molecular models enable the distinction of explicit signals such as scaffolds? The answer appears to be no. While vanilla 2D-3D contrastive pretraining shows clustering phenomena, it lacks fine-grained semantic information, as evidenced by the visualization of molecular representation in Fig. 5(a). To address this, we integrate molecular fingerprints as a prior. Equipped with Knowledge-Guided Prototypical Clustering, our model can distinguish scaffolds without explicit supervision, as shown in Fig. 5(b). Although recent models like MoleBlend and those using Transformer-M as a backbone [87] have not adopted 2D-3D molecular contrastive learning, our approach fills this gap. Regarding node-level pretext tasks, InfoMotif and Denoising serve as complementary tasks to obtain high-quality embeddings at the node level, as demonstrated in Table 4. The implicit 3D geometry’s effectiveness is highlighted by the fact that using only 2D information can produce results identical to those using 3D information, as shown in Fig. 4. Our approach aims to align the knowledge gained from both 2D and 3D modalities through contrastive learning, rather than treating it as an end in itself. The pretext tasks collectively contribute to the comprehensive pretraining of our MolGT framework.

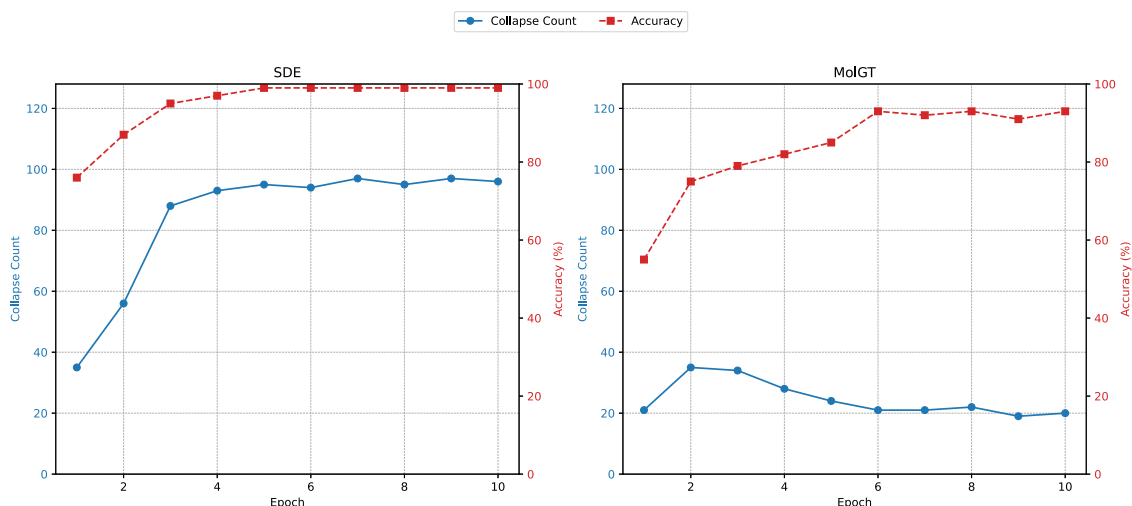


Fig. D.8. The plots depict the number of collapsed dimensions and the accuracy of the contrastive learning task over epochs. The x-axis represents the epochs, while the left y-axis shows the number of collapsed dimensions, and the right y-axis indicates the accuracy percentage. We count the collapsed dimensions based on whether their singular values collapse to zero.

Data availability

Data will be made available on request.

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